UNIVERSIDAD NACIONAL DE COLOMBIA

Respuestas - Taller de Calculo Integral III

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1 Integracion por Partes

1. $\int xe^x\,dx = e^x(x-1) + C.$
2. $\int x\ln x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C.$
3. $\int x^2\cos x\,dx = x^2\sin x + 2x\cos x - 2\sin x + C.$
4. $\int e^{2x}\sin x\,dx = \frac{e^{2x}}{5}(2\sin x - \cos x) + C.$
5. $\int \arctan x\,dx = x\arctan x - \frac{1}{2}\ln(1+x^2) + C.$

2 Fracciones Parciales

1. $\int \frac{3x+5}{x^2-1}\,dx = 4\ln|x-1| - \ln|x+1| + C.$
2. $\int \frac{2x^2+3x+1}{x(x+1)(x+2)}\,dx = \frac{1}{2}\ln|x| + \frac{3}{2}\ln|x+2| + C.$
3. $\int \frac{dx}{x^2+5x+6} = \ln|x+2| - \ln|x+3| + C.$
4. $\int \frac{x+4}{(x-1)^2}\,dx = \ln|x-1| - \frac{5}{x-1} + C.$
5. $\int \frac{4x}{x^2(x^2+1)}\,dx = 4\ln|x| - 2\ln(x^2+1) + C.$

3 Integrales Trigonometricas y Sustitucion Trigonometrica

1. $$\int \sin^2 x\cos^2 x\,dx = \frac{x}{8} - \frac{\sin(4x)}{32} + C.$$
2. $$\int \frac{dx}{\sqrt{9-x^2}} = \arcsin\left(\frac{x}{3}\right) + C.$$
3. $$\int \sqrt{16-x^2}\,dx = \frac{x}{2}\sqrt{16-x^2} + 8\arcsin\left(\frac{x}{4}\right) + C.$$

4.
$$\int \frac{x^2}{\sqrt{x^2+4}}\,dx = \frac{x}{2}\sqrt{x^2+4} - 2\ln\left|x+\sqrt{x^2+4}\right| + C.$$

5. Para $x > 1$:
$$\int \frac{dx}{x^2\sqrt{x^2-1}} = \frac{\sqrt{x^2-1}}{x} + C.$$

4 Integrales Impropias

1.
$$\int_1^\infty \frac{1}{x^p}\,dx$$
converge si y solo si $p > 1$, y en ese caso su valor es
$$\frac{1}{p-1}.$$

2. $\int_0^1 \frac{1}{\sqrt{x}}\,dx = 2.$

3. $\int_0^\infty e^{-2x}\,dx = \frac{1}{2}.$

4. $\int_{-\infty}^\infty \frac{1}{x^2+1}\,dx = \pi.$

5 Aplicaciones Integrales

1. Si $v(t) = 3t^2 - 6t + 2$ y $s(0) = 5$:
$$s(t) = \int (3t^2 - 6t + 2)\,dt = t^3 - 3t^2 + 2t + C, \quad C = 5,$$
$$s(t) = t^3 - 3t^2 + 2t + 5.$$

2. Trabajo en el resorte:
$$W = \int_0^{0.2} 120x\,dx = 60x^2\Big|_0^{0.2} = 2.4 \text{ J}.$$

3. Si $C'(q) = 0.04q + 15$ y $C(0) = 800$:
$$C(q) = 0.02q^2 + 15q + 800.$$